

INITIAL TOWER TECHNICAL ASSESSMENT

Project Valkyrie: EMEA Datacenter Exit & Hybrid Cloud Transformation

AETHELGARD LIFE SCIENCES

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1. Executive Summary

Aethelgard Life Sciences is embarking on Project Valkyrie, a strategic initiative to exit the primary EMEA datacenters located in Basel (Switzerland) and Frankfurt (Germany). This initiative is driven by impending hardware end-of-life (EOL) cycles across the legacy footprint, rigid capacity constraints limiting R&D High-Performance Computing (HPC) capabilities, and absolute deadlines imposed by facility lease expirations in Q4 2027.

The Initial Tower Technical Assessment (ITTA) evaluates the current state of Aethelgard's infrastructure, network, security, and application landscapes. The target state vision encompasses a hybrid architecture, heavily leveraging Amazon Web Services (AWS) as the primary cloud service provider for modernized and re-platformed workloads. This will be seamlessly integrated with an Equinix colocation model engineered to support latency-sensitive Manufacturing Execution Systems (MES) and legacy AS/400 systems that cannot be safely refactored within the strict 18-month timeline.

This document outlines the high-level technical assessment across nine critical IT towers. It details operational dependencies, stringent GxP compliance requirements (including FDA 21 CFR Part 11), and tower-specific modernization recommendations to ensure uninterrupted operational continuity, robust security posture, and regulatory adherence throughout the transition.

2. Business Context

2.1 Demand Overview

Following the strategic acquisition of BioGenix Innovations in 2025, Aethelgard's IT footprint expanded significantly, inheriting technical debt and disparate legacy environments. The combined enterprise now supports over 42,000 employees globally. The Basel and Frankfurt datacenters currently host approximately 60% of Aethelgard's core commercial, clinical, and ERP workloads. The combined physical estate encompasses over 1,400 virtual machines (VMs), 6 PB of active storage, and 650 distinct business applications spanning manufacturing, HR, finance, and specialized clinical trials.

2.2 Objectives

- **Infrastructure Agility:** Shift from a static, CAPEX-heavy data center model to a highly scalable OPEX cloud model, specifically enabling dynamic scaling for genomics sequencing and R&D data pipelines.
- **Risk Mitigation:** Vacate the Basel and Frankfurt facilities prior to the November 2027 lease termination to avoid multi-million-euro penalty clauses.
- **Modernization:** Retire 15% of the legacy application portfolio, re-platform 45% utilizing PaaS services (e.g., AWS RDS, Amazon EKS), and rehost the remaining 40% (IaaS) utilizing block-level replication.
- **Regulatory Compliance:** Maintain uninterrupted adherence to EMA guidelines, FDA 21 CFR Part 11, HIPAA, and GDPR regulations throughout the entire migration lifecycle.

2.3 Scope

In Scope	Out of Scope
• All workloads currently hosted in the Basel and Frankfurt physical datacenters. • Global MPLS to SD-WAN network migration for interconnected clinical sites. • Identity & Access Management (IAM) modernization and cloud federation. • R&D HPC compute environments and large-scale data storage arrays. • SAP ECC to S/4HANA infrastructure groundwork (pre-migration foundational setup).	• End-User Computing (EUC) including VDI, laptops, and mobile device management. • APAC region datacenters (Singapore, Tokyo) and associated localized networks. • Office 365 and existing SaaS application configurations. • Physical security and access control systems of manufacturing sites.

3. Assumptions & Dependencies

Critical Dependencies:

- **Network Provisioning:** AWS Direct Connect (Dual 10Gbps dedicated circuits) must be provisioned and operational at the Equinix edge locations by Q3 2026. Delays in circuit delivery will directly impact Wave 1 migration schedules and invalidate latency testing.
- **Business Validation:** Application Owners must commit up to 20% resource allocation for User Acceptance Testing (UAT) and GxP validation sign-offs during their respective migration waves. Lack of UAT sign-off blocks cutover.
- **CMDB Accuracy:** Current ServiceNow CMDB completeness is assumed to be approximately 75%. Deep discovery utilizing Flexera/Device42 will run in Month 1 to resolve the remaining 25% dependency mappings and blind spots.

Assumptions: The transition budget of €35M is fully approved, capitalized, and accessible. Existing enterprise software licenses (Oracle, Microsoft, SAP) are verified for Bring-Your-Own-License (BYOL) cloud mobility. No major functional changes, version upgrades, or net-new feature developments will occur for applications classified as "Rehost" during the migration window (Code Freeze period).

4. High-Level State Assessment

4.1 Current State Architecture

The existing environment relies on a traditional three-tier architecture utilizing Cisco ACI for network virtualization, Dell/EMC VMAX for Tier 1 block storage, and VMware vSphere 7.0 for compute virtualization. The database

landscape is heavily fragmented and severely reliant on Oracle Enterprise 19c deployed on bare metal, leading to exorbitant licensing costs. Furthermore, there are older instances of Microsoft SQL Server (2012/2014) requiring extended security updates. Perimeter security relies on legacy Palo Alto hardware firewalls with minimal intra-VLAN segmentation (flat network topology), creating a substantial risk for lateral movement in the event of a breach.

4.2 Target State Vision

The target architecture will pivot to an AWS Multi-Account Landing Zone based on AWS Control Tower, rigorously segmented by environment (Dev, QA, Prod) and business unit (Commercial, R&D, Corporate). Compute platforms will transition away from hypervisor-based VMs to Amazon EC2, AWS Fargate, and Amazon EKS where applicable. Oracle workloads will aggressively migrate to Amazon Aurora PostgreSQL (where schema conversion is viable via SCT) or Amazon RDS for Oracle. The network architecture will abandon the traditional perimeter model, adopting a Zero Trust network access (ZTNA) model leveraging Zscaler Private Access (ZPA) alongside an AWS Transit Gateway for site-to-cloud scalable connectivity.

5. Tower-Level Assessments

5.1 Cloud & Migration Strategy

Current State: The current cloud footprint is minimal, primarily consisting of isolated, shadow-IT AWS accounts utilized sporadically by R&D data scientists without centralized governance, unified networking, or billing consolidation.

Target State: A fully governed, enterprise-grade AWS environment. Deployment of a tailored Landing Zone utilizing AWS Control Tower. Strict implementation of Infrastructure as Code (IaC) via HashiCorp Terraform Enterprise for all cloud resource provisioning.

Migration Considerations:

- **Tooling:** AWS Application Migration Service (MGN) will be utilized for continuous, block-level replication of Rehost (lift-and-shift) workloads to minimize downtime.
- **Wave Planning:** Migration waves will be formulated based on tight affinity groups (highly interdependent applications) identified via discovery tools. This prevents "split-brain" latency issues where an application sits in AWS while its database remains on-premises.

Risk Identified	Severity	Mitigation Strategy
Legacy OS compatibility (Windows Server 2008/2012) lack of support in AWS AMIs.	High	Leverage AWS End-of-Support Migration Program (EMP) for packaging, or isolate in hardened VPCs with restricted routing if OS upgrade is impossible.
IP overlap between current on-premises DCs and default AWS VPC CIDR blocks.	Medium	Network Architecture team to design a comprehensive IPAM strategy utilizing AWS IPAM prior to initial Landing Zone deployment.

5.2 Security & Identity

Current State: The environment relies on a legacy perimeter-centric security model. Authentication is handled by on-premises Microsoft Active Directory domains with fragmented Single Sign-On (SSO) integrations. There is a distinct lack of East-West traffic inspection within the data centers.

Target State: Comprehensive Zero Trust architecture. Okta will serve as the centralized Identity Provider (IdP), natively integrated with AWS IAM Identity Center for both programmatic and console access. Zscaler ZPA will replace legacy Cisco AnyConnect VPNs, providing application-level access rather than network-level access.

Compliance Impact: As a pharmaceutical entity, Aethelgard must ensure all AWS services utilized for regulated workloads are HIPAA eligible and configured in accordance with stringent GxP guidelines. Core security services including AWS CloudTrail, AWS Config, and Amazon GuardDuty must be enabled across all member accounts, with immutable logs forwarded to a locked, centralized Security tooling account.

5.3 Infrastructure & Hosting

Current State: Aging Dell PowerEdge blade chassis and EMC storage arrays are scheduled to reach physical EOL in early 2027. The landscape includes dedicated AS/400 mainframes critical for legacy supply chain logistics and routing.

Hosting Strategy:

- **Cloud (85%):** Standard corporate workloads, web tiers, application servers, and modifiable relational databases will transition completely to AWS.
- **Colocation (15%):** The AS/400 mainframe and localized Manufacturing Execution Systems (MES) managing physical automated manufacturing lines require ultra-low latency (<5ms). These highly specific workloads will be physically migrated to an Equinix colocation facility in Frankfurt via secure transport, connected directly to AWS via Direct Connect for reporting and data aggregation.

5.4 Network & Connectivity

Connectivity Strategy: Establish a highly resilient AWS Transit Gateway (TGW) architecture to act as the centralized cloud router. Dual AWS Direct Connect (DX) links will be established from the Equinix edge back to the AWS eu-central-1 (Frankfurt) region, ensuring redundancy and sub-millisecond cloud-to-colo latency.

Segmentation: Implementation of strict micro-segmentation utilizing AWS Security Groups and Network Access Control Lists (NACLs). A default-deny posture will be adopted. All cross-environment traffic (e.g., Dev attempting to reach Prod) will be explicitly dropped and logged for SIEM analysis.

5.5 Database Architecture

Current State: Over 180 massive Oracle Enterprise databases, consuming high licensing costs due to core-based pricing models. Hardware is heavily over-provisioned simply to support peak periodic loads, leading to massive inefficiencies.

Data Migration Strategy:

- **Refactoring & Optimization:** The AWS Schema Conversion Tool (SCT) and Database Migration Service (DMS) will be deployed. The technical objective is to target 30% of standard Oracle workloads for refactoring to Amazon Aurora PostgreSQL to drastically reduce enterprise licensing TCO.
- **Heavy Storage Transfer:** Petabyte-scale R&D genomic data archives will be transferred utilizing AWS Snowball Edge hardware devices to bypass unfeasible network bandwidth constraints, landing directly in Amazon S3 Glacier Flexible Retrieval.

Backup/DR: Transition from legacy Commvault physical tape backups to AWS Backup. Establish aggressive RPO/RTO tiers mapped directly to business criticality (e.g., Tier 1 MES systems mandate RPO of 15 mins, RTO of 1 hour, achievable via Multi-AZ RDS deployments).

5.6 Applications & Integration

Modernization Approach:

- **Containerization:** Stateless internal web applications and custom microservices will be repackaged into Docker containers and orchestrated globally via Amazon EKS.
- **Integration Modernization:** The legacy monolithic Enterprise Service Bus (Microsoft BizTalk) will be systematically phased out. APIs will be exposed, secured, and managed via MuleSoft CloudHub, operating as the central, modernized enterprise API gateway.

Integration Challenge	Mitigation Approach
Hardcoded IP addresses deeply embedded in legacy application configuration files.	Mandatory code remediation phase enforced during Wave 0. Implement Amazon Route 53 private hosted zones for internal DNS resolution and enforce FQDN usage.
Network latency between split-tier applications during the hybrid migration phase.	Strict dependency affinity grouping. The application tier and its corresponding database tier MUST migrate in the exact same scheduled weekend downtime window (Move Groups).

5.7 Governance & Compliance

Risk Controls & Audit: Migrating to a public cloud introduces structural changes to validated systems (GxP). A modernized Computer Systems Validation (CSV) framework adapted for cloud infrastructure (Continuous Compliance) must be finalized and approved by the Quality Assurance board prior to Wave 1.

Compliance Mapping:

- **21 CFR Part 11:** Electronic signatures and digital audit trails must remain independently verifiable. AWS CloudTrail and application logs will be stored in a WORM (Write Once Read Many) state utilizing S3 Object Lock.
- **Data Sovereignty:** Due to stringent GDPR constraints and the Swiss Data Protection Act, no primary or backup data may reside outside the European footprint. Disaster Recovery architecture will replicate data seamlessly from eu-central-1 (Frankfurt) to eu-west-1 (Ireland) or eu-west-3 (Paris) exclusively.

5.8 Monitoring & Observability

Logging & SIEM: Splunk Enterprise Security will remain the core SIEM platform. Splunk forwarders and Amazon Kinesis Data Firehose integrations will securely ingest VPC Flow Logs, CloudTrail, GuardDuty findings, and OS-level security events.

Alerting Strategy: Transition from traditional infrastructure-centric alerts (e.g., "CPU utilization > 90%") to modern application-centric observability utilizing Datadog. Engineering teams will define comprehensive Service Level Indicators (SLIs) for key business transactions to monitor true user experience.

6. High-Level Transition Approach

Project Valkyrie will be strictly executed in a phased, wave-based approach across 18 months, ensuring continuous risk management, rapid feedback loops, and minimal disruption to critical manufacturing production lines.

- **Phase 1: Foundation (Months 1-3):** Architect and deploy the AWS Landing Zone, establish Direct Connect networking, deploy foundational IAM Okta integrations, and finalize deep dependency discovery (Flexera).
- **Phase 2: Pilot & Non-Prod (Months 4-6):** Migrate Development and QA environments. Refine Runbooks, execute rigorous performance and latency testing, and validate IaC deployment pipelines.
- **Phase 3: Production Waves (Months 7-15):** Execute tiered migration waves. Critical path applications (ERP, MES dependencies, clinical trial databases) are scheduled for Month 12 to ensure environmental stabilization well prior to the absolute DC closure date.
- **Phase 4: Decommissioning (Months 16-18):** Physical hardware wiping (NIST 800-88 compliance), facility lease handover procedures, and finalization of legacy vendor contract terminations.

7. Recommendations Summary

Based on the initial assessment, the Enterprise Architecture Review Board highly recommends the following immediate actions to secure the project timeline and mitigate risk:

1. **Accelerate Network Procurement:** Immediately execute Master Service Agreements (MSAs) for AWS Direct Connect and Equinix colocation space, as regional telecom provisioning lead times currently average 90-120 days.
2. **Establish Cloud Center of Excellence (CCoE):** Formally charter a dedicated CCoE consisting of Cloud Architects, Cloud Security Engineers, and Quality Assurance (QA) personnel to govern the ongoing GxP validation of cloud components.
3. **Finalize App Disposition:** Mandate business unit executive sign-off on the initial 6Rs (Rehost, Replatform, Refactor, Repurchase, Retire, Retain) disposition matrix by the end of Q2 2026.
4. **Initiate Oracle Optimization:** Engage database specialists to immediately begin schema analysis for the top 20 highest-cost Oracle databases to validate Amazon Aurora PostgreSQL target viability, maximizing ROI early in the project.

Approval for subsequent Low-Level Design (LLD) phases and budget release is contingent upon the formal endorsement of this ITTA by the Global IT Steering Committee.